Effect of Infrared Sensor Configuration on Leader-Follower Tracking Accuracy using the Pololu 3Pi+

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***Abstract*—This paper investigates the impact of infrared sensor configuration on leader-follower tracking accuracy of two Pololu 3Pi+ robots. The performance of forward-facing bump sensors and downward-facing line sensors was evaluated under varying initial longitudinal separations, lateral offsets, and leader trajectories. Results showed that the two sensor configurations provided distinct operating regions. Line sensors offered wider lateral coverage but a shorter effective forward range, while bump sensors provided a narrower field of view. In both straight-line and arcing trials, the bump sensor follower achieved lower mean RMS radial tracking error than the line sensor follower, with an average improvement of 13.3 mm across all initial longitudinal separations. These findings indicate that sensing configuration and geometry significantly influence tracking performance, and that for the conditions considered, forward-facing bump sensors provide a more accurate solution.**

I. Introduction

Coordinated motion between multiple autonomous robots is a key capability in modern robotics, enabling applications in areas such as agriculture, logistics, and swarm systems [1], [2]. Leader-follower architectures provide a practical approach to such coordination, where a designated leader defines the trajectory and follower robots maintain a relative position, reducing system complexity while enabling structured group behaviour.

The effectiveness of this approach depends on the ability of the follower to accurately estimate the relative position of the leader using onboard sensing. While high-fidelity sensing methods such as vision-based and LiDAR can provide accurate measurements, they introduce significant cost and computational requirements. In contrast, infrared (IR) sensing offers a lightweight and low-cost alternative suitable for compact robotic platforms, but its performance is strongly influenced by sensor geometry/configuration, signal interpretation, and environmental conditions.

Despite its widespread use, there is limited quantitative understanding of how IR sensor configuration affects leader-follower tracking accuracy, particularly under variations in spatial offset and motion. On platforms such as the Pololu 3Pi+ [3], used in the present investigation, multiple IR sensing mechanisms are available, including the forward-facing bump sensors and the downward-facing line sensors; fundamentally differing in their field of view and signal characteristics. However, their relative effectiveness for leader detection and tracking has not yet been systematically evaluated.

This project addresses this gap by experimentally comparing two follower implementations based on these sensing configurations. Performance is evaluated under controlled variations in longitudinal separation, lateral offset, and trajectory type, and within a common operating region (X_0: 100–120 mm, Y_0: within $\pm$20mm) selected to ensure both sensor configurations remained within a comparable sensing range. Preliminary signal-mapping experiments are used to characterise the sensing workspace of each configuration, informing both the experimental design and the hypotheses tested. The overall contribution of this work is a systematic, experimentally grounded evaluation of low-cost IR sensing strategies for leader–follower robotics, with particular emphasis on the impact of sensing geometry on tracking accuracy.

A. Hypothesis Statement

> *As the initial longitudinal separation, lateral offset, and trajectory curvature increase, tracking error in leader-follower operation will increase. Specifically, bump sensor-based tracking is expected to achieve lower longitudinal error at larger separations, whereas line sensor-based tracking is expected to achieve lower lateral and overall positional error under lateral offsets and curved trajectories.*

Increasing longitudinal separation reduces the intensity of the IR signal received by the follower, degrading the reliability of distance estimation. Conversely, lateral offsets and curved trajectories place greater emphasis on accurate bearing estimation and continuous directional correction. The bump sensors observe the emitted IR signal from the leader along a narrower, more direct axis, which is expected to preserve distance-related information more effectively at larger longitudinal separations. In contrast, the circumferential arrangement of the line sensors provides a wider lateral sampling region, enabling improved detection of angular deviations and more robust tracking under laterally offset and curved motion.

II. Implementation

A. Leader-Follower Control Architecture

The experiment tested a leader-follower architecture utilising two sensor configurations, bump and line. Both implementations used the same overall control architecture. A leader robot followed a predefined trajectory while emitting an IR beacon, and a follower robot attempted to maintain a desired relative position using IR sensing feedback as shown in Figure 1.

The follower architecture consisted of wheel-speed estimation, odometry-based pose estimation, and sensor-based longitudinal and lateral correction. The two implementations differed only in the sensing configuration and

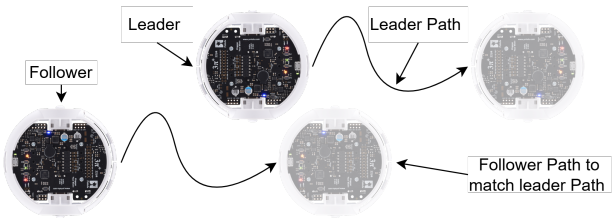


Fig. 1: Illustrative example of Leader-Follower behaviour with the follower maintaining initial separation to the leader throughout motion

in the method used to derive signal measures from the IR readings. At the lowest level, PID wheel controllers generated left and right motor PWM commands from the difference between the desired wheel-speed demands and the measured wheel speeds, where wheel speed was estimated from encoder count change over elapsed time in counts/ms as per [3].

Above this level, the follower computed two higher-level correction terms from the IR measurements. The first was a longitudinal controller, which generated a forward demand based on the difference between the measured total beacon signal and a previously sampled target signal level, thereby regulating the follower's distance relative to the leader. The second was a lateral controller, which generated a steering correction from the difference between the left and right sensor signals relative to the initially sampled bearing offset. These two terms were then combined through a differential-drive formulation to generate the left and right wheel-speed demands, as shown in Equation 1.

$$V_L = V_f + \omega, \qquad V_R = V_f - \omega \tag{1}$$

where V_f is the forward velocity demand and ω is the steering correction derived from sensor asymmetry. These terms are described further in Sections II-D and II-E.

This architecture was used to isolate the independent variable of interest: the sensor configuration. Any observed performance differences are therefore intended to arise from how the two sensing arrangements provide IR information to the same underlying follower control structure. The controller gains used in the leader and follower implementations are summarised in Table I.

TABLE I: Controller gains used in the leader and follower systems, with K_p, K_i, and K_d being proportional, integral, and derivative (PID) gains respectively.

Agent	Controller	K_p	K_i	K_d
Follower	Left wheel speed	15.0	0.1	0.0
Follower	Right wheel speed	15.0	0.1	0.0
Follower	Distance (bump)	0.03	–	–
Follower	Bearing (bump)	0.0050	–	–
Follower	Distance (line)	0.01	–	–
Follower	Bearing (line)	0.12	–	–
Leader	Left wheel speed	60.0	0.02	0.001
Leader	Right wheel speed	60.0	0.02	0.001
Leader	Heading	2.0	–	–

Controller gains were tuned empirically by minimising oscillation at X_0=100mm, and Y_0=0mm, independently for each sensor configuration.

B. Sensor Operation

The follower robot can utilise either the bump IR sensors or line IR sensors to detect the IR beacon emitted by the designated leader. In the leader-follower arrangement used here, the leader emits IR from its bump/line emitters while reversing. This allows the bump sensors located at the front of the follower to observe the beacon directly, whereas the line sensors observe the signal indirectly through reflection from the test surface. In this experiment, the sensors were read using a digital read approach, where reading magnitude and thus IR intensity was inferred from a charge-discharge timing measurement, stronger received IR corresponded to a shorter discharge time.

Due to the two sensor types having different geometries, each is assumed to display different sensing characteristics. The bump sensors are oriented forward and thus are expected to have a narrow field of view and longer usable forward range compared to the line sensors. The line sensors are reflectance-based and located circumferentially around the front of the follower. As such they are suspected to operate over a wider lateral sensing region, but their measurements depend more strongly on the reflective properties of the surface and on discharge timing. These physical differences are expected to influence both the available working area and the quality of the distance and bearing signals formed by the controller.

C. Signal Normalisation and Baseline Calibration

The raw IR readings were calibrated to the environment and normalised relative to a baseline measurement captured before activation of the leader beacon. This reduced the effect of ambient lighting and static sensor offsets.

For the bump-sensor implementation, the total signal was formed from the sum of the left and right bump-sensor signals. For the line-sensor implementation, the two left-most and two right-most sensors were aggregated into left and right groups respectively, while the centre sensor was retained separately where required. The grouped signals were then summed to form the total received signal.

This baseline-relative formulation reduced sensitivity to static offsets caused by ambient illumination and surface reflectance, and allowed the initial sampling phase to define a reference operating condition for both distance and bearing. To further reduce environmental variation, all experiments were conducted in rooms with minimal external light sources and a low-pass filter was applied to the measured readings to mitigate sensor noise variations.

D. Longitudinal Control

The longitudinal controller regulated the distance between the follower and leader using the total IR signal, S_T. A target signal level was obtained during an initial sampling phase in which the follower was positioned at the desired separation from the leader. The forward velocity demand was then computed as equation 2.

$$V_f = k_V(S_{T,target} - S_T) \tag{2}$$

where $S_{T,target}$ is the sampled reference signal and K_V is the proportional distance-control gain. A lowpass filter was applied to S_T and a dead band to $S_{T,target}$ to reduce actuation caused by small signal oscillations in the follower's received signal when near the desired separation.

This approach assumes that received IR intensity varies consistently with relative separation, allowing the follower

to maintain approximate longitudinal spacing without requiring an explicit distance measurement.

E. Lateral Control

Lateral alignment is achieved using a simple bearing signal derived from the difference between left and right sensors shown in equation 3

$$B = \frac{L - R}{L + R} \tag{3}$$

Where B is the normalised bearing signal with dimensionless units, and a range of -1 to +1, and L and R are the left and right grouped IR sensor readings respectively. This normalised difference provides a measure of the angular offset between the follower and leader robots. During motion a steering condition is then applied, shown in equation 4 to maintain desired constant bearing and thus lateral offset.

$$\omega = K_B(B - B_{target}) \tag{4}$$

Where ω is the steering correction, K_B is the proportional bearing gain, and B_{target} is the desired bearing signal. To improve stability, a low-pass filter is applied to the bearing signal and a deadband is used to ignore small sensor noise variations.

F. State Machine and Data Acquisition

The follower operated using a finite state machine in order to synchronise its behaviour with that of the leader. The overall procedure is summarised in Algorithm 1. First, the follower measured a baseline IR signal while the leader beacon was absent. The follower then sampled the beacon while the leader was held at the desired starting position, allowing reference values for total signal and bearing to be established. After sampling, the follower waited for the leader beacon to be switched off and then on again before entering closed-loop tracking mode.

Algorithm 1 Follower control procedure

```
if S_T > S_threshold then
    reset lost-signal timer
    if new sensor reading is available then
        measure and filter current total signal, T
        measure and filter current bearing signal, B
        V_f < −K_f(S_T,target − S_T)
        ω = K_b(B − B_target)
        compute wheel demands, V_(L,R) ← V_f, ω
    end if
    update wheel-speed controllers
    log positional data, X, Y
else
    maintain previous V_L and V_R
    if lost-signal timer not started then
        start lost-signal timer
    end if
    if elapsed lost-signal time > T_timeout then
        stop robot
        stop logging
        terminate trial
    end if
end if
```

During tracking, the follower continuously updated its longitudinal and lateral control demands from the measured IR signals. When the leader completed its run and the beacon signal dropped below a threshold for longer than a predefined timeout, the follower terminated the trial and stopped. This event-driven procedure ensured that both robots began motion under consistent conditions and reduced timing misalignment between recorded leader and follower trajectories.

Each robot logged odometry-based positional data with timestamps at a fixed interval of 50 ms. Data were stored onboard in SRAM and retrieved after the trial, allowing experiments to be run without a permanent serial connection. The logged variables were the x and y positions represented as scaled 16-bit integers in order to reduce memory use while preserving sufficient resolution for later analysis. Data were recorded over the full duration of each following run, which was approximately 8 s for the straight-line task and 10 s for the curved-path task. The leader and follower datasets were then aligned in post-processing as discussed in Section III-D.

III. Experiment Methodology

A. Experimental Setup

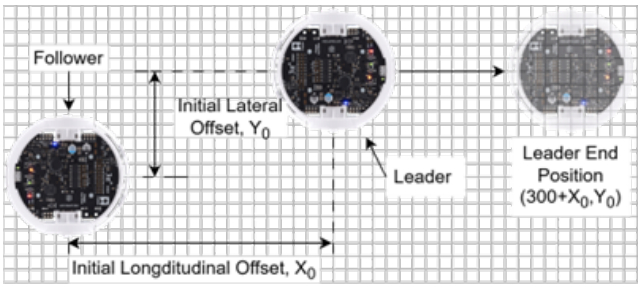
The experimental setup is illustrated in Figures 2a and 2b for the straight and curved path setups respectively. An A3 printed test surface containing a reference grid was placed on a flat, level surface. The reference grid consisted of lines for the alignment of both the leader and follower at 5 mm intervals in both the lateral and longitudinal directions. This flat test surface ensured consistency across both paths and the reference lines ensured that all tests were initiated with consistent starting positions. Additionally the grid printing facilitated much easier odometry calibration, with the lines made intentionally faint as to limit the disruption of the line sensor's readings.

B. Experimental Procedure

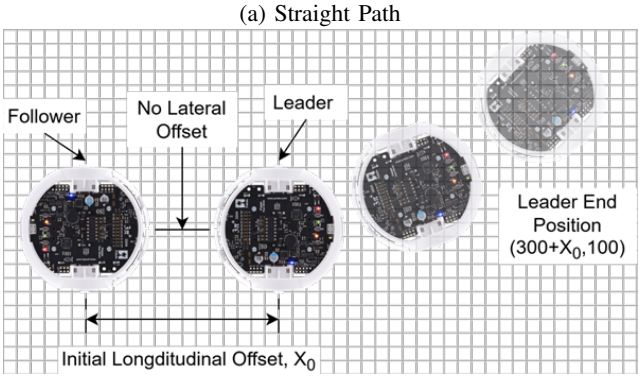
The main study used a common experimental procedure for two follower sensor configurations across two trajectory types: a straight-line path and a curved path. In addition, separate sensor-characterisation experiments were carried out in advance in order to define a common operating region for the main tests.

1) General Procedure: The general procedure is summarised by the flow diagram in Figure 3. At the beginning of each test, the leader and follower were positioned at their designated locations on the test grid, facing in opposite directions. The follower was powered and allowed to remain idle briefly so that ambient IR conditions could be captured during baseline calibration. After calibration, the leader beacon was activated using Button A and the follower sampling phase was initiated using its own Button A, recording target distance and bearing signals.

Once sampling was complete, the leader beacon was switched off and the follower entered a waiting state. The trial was then started by pressing Button B on the leader, which initiated leader motion after an 800 ms pre-start delay. This delay allowed the follower to detect reactivation of the beacon and enter tracking mode under consistent conditions. After the leader completed its run and the



(a) Straight Path



(b) Curved Path

Fig. 2: Experimental Setup with initial longitudinal, X_0 and lateral, Y_0 offsets and highlighting leader end position for both the a) straight path and b) curved path tasks.

follower terminated tracking, both robots were connected to a PC via serial and their logged positional data were exported for post-processing. Each trial was repeated 10 times for each sensor configuration and standard deviations taken from the results.

2) Sensor Characterisation: To characterise the sensing workspace of each configuration and to define an appropriate common testing region, a secondary test was performed in which the total IR signal observed by the follower was measured while the leader was placed at different relative positions with its beacon active. The leader was positioned over the range $x = 90$-150 mm and $y = -50$-50 mm in 10 mm increments, while the follower remained stationary. Three measurements were taken at each position for both sensing configurations and averaged to produce the results shown in section IV.

C. Discussion of Variables

- **Independent Variable**: The leader trajectory type was varied in order to compare straight-line and curved-path following. The initial longitudinal and lateral offsets between the leader and follower were also varied within a operating region common to both sensing configurations. In the straight-line and curved-path tests, the initial longitudinal offset, X_0, was varied over $100 - 120\ mm$ in $5\ mm$ increments for all trials, while the initial lateral offset, Y_0, was varied over $-20 - 20\ mm$ in $10\ mm$ increments for the straight-line trials.
- **Dependent Variable(s)**: The primary dependent variables were the leader and follower positions as functions of time, obtained from onboard odometry logs. A secondary dependent variable was the total IR signal measured during sensor-characterisation experiments.

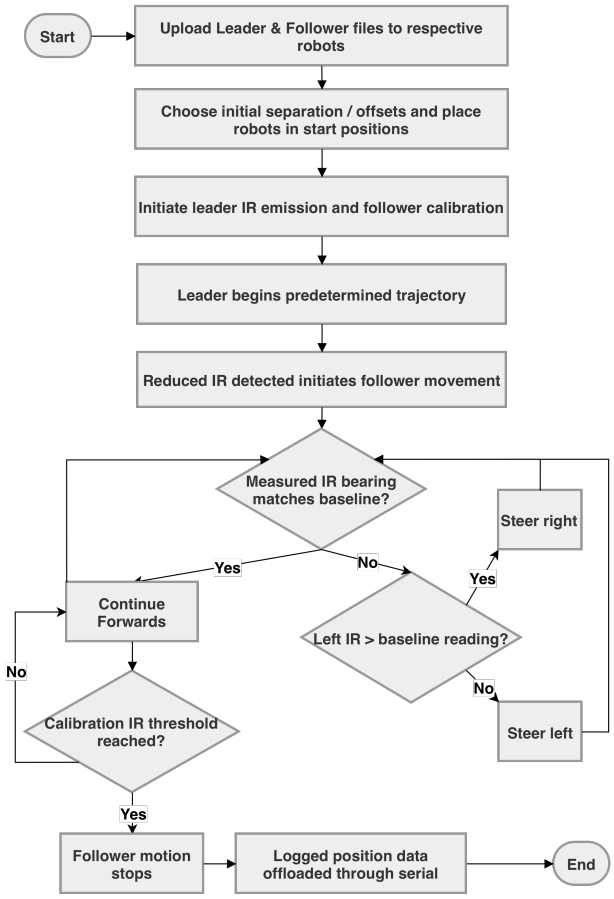


Fig. 3: Flow diagram summary of experimental procedure

- **Controlled Variables**: Several variables were controlled to improve repeatability. The same pair of robots were used throughout all experiments with fixed leader and follower roles, both robots had their outer casing removed for all tests to enable reading of the bump emitters by the follower and minimise disruption to IR during line sensor trials. The test surface was kept constant in order to minimise variation in line-sensor IR reflectance. The same logging procedure, baseline calibration method, and underlying control structure were used across sensing configurations. Additionally, before each set of tests were performed, it was ensured that the batteries for both robots were fully charged. In all cases, the robots were aligned parallel to their initial direction of travel to minimise initial bearing error that could propagate into their odometry.

D. Discussion of Metrics

The first quantity analysed was the total received IR signal measured over a grid of leader positions. This quantity was calculated using Equations 5 and 6 and was used to characterise the sensing workspace of each configuration, as well as to define a common operating region for the main experiments. A sensing configuration that maintained a stronger and more spatially extensive signal was expected to support more robust tracking performance.

$$S_j = \min\left(100,\ \max\left(0,\ 100\frac{B_j - R_j}{B_j}\right)\right) \tag{5}$$

$$S_{\text{Total}} = S_L + S_R + S_C \tag{6}$$

where R_L, R_R, and R_C are the raw combined left, combined right, and centre sensor readings, respectively; B_L, B_R, and B_C are the corresponding baseline readings measured during calibration; and S_L, S_R, and S_C are the corresponding normalised sensor signals. For the bump sensor configuration, the total signal reduces to the sum of the available sensor signals only.

The main tracking metrics were the root-mean-square (RMS) radial error. After each run, the leader and follower odometry datasets were trimmed to remove pre-start and beacon-loss timeout periods, aligned in time, and offset according to the initial separation. The RMS radial error was then calculated using equation 7 combining the calculated Longitudinal and lateral deviations throughout the trajectory from equation 8.

$$\text{RMS}_r = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(e_x^2(t_i) + e_y^2(t_i)\right)} \tag{7}$$

Where,

$$e_k(t_i) = (k_L(t_i) - k_F(t_i)) \tag{8}$$

where $k \in \{x, y\}$ denotes the coordinate direction, $k_L(t_i)$ and $k_F(t_i)$ are the leader and follower positions at sample t_i, N is the total number of aligned samples, and RMS_r is the root-mean-square radial error.

Smaller RMS errors indicate that the follower maintained closer adherence to the desired relative position. These metrics were therefore used to compare the effectiveness of the bump-sensor and line-sensor implementations under varying offsets and trajectory conditions.

IV. Results

A. Sensor Characterisation

Figure 4 shows the measured relative signal strength for both sensing configurations over the sampled leader-offset grid. Peak signal levels within the effective operating regions were comparable, whereas spatial distributions of usable signals differed substantially.

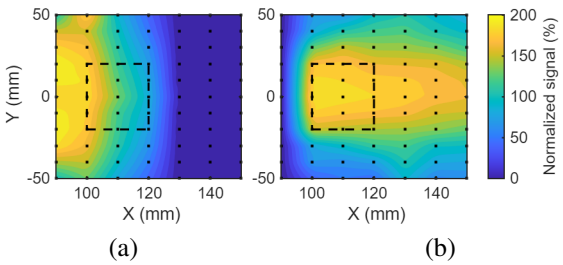


Fig. 4: Measured relative signal strength for varying leader offsets using (a) line sensors and (b) bump sensors. Black markers indicate sampled leader positions.

The line-sensor configuration exhibited a broader lateral operating region, with signal strength degrading approximately radially as longitudinal and lateral offset increased. However, the signal became indistinguishable from the ambient baseline beyond an $X = 130$ mm, indicating a limited forward range.

The bump-sensor configuration defines a longer direct line-of-sight operating range but over a narrower lateral field of view, with a more symmetric signal decay as lateral offset increases. At $X < 100$ mm, the bump-sensor signal was limited by the physical arrangement of emitters and receivers, producing a proximal region in which little usable signal was detected.

These results show that the two sensing configurations provide fundamentally different sensing workspaces rather than simply different signal magnitudes. Based on these measurements, and preliminary testing revealing a minimum threshold for repeatable follower behaviour to be signal of approximately 80%, a common testing region was selected for the main experiments to ensure equitable comparison of tracking performance.

B. Arcing Path Trial

Figure 5 shows the mean RMS error from the curved path trial, overlaid with derived uncertainty from the variation between the 10 tests. Radial error rose with X_0 for both sensor configurations. The bump-sensor error increased by 63.2% from 87.63 mm at $X_0 = 100$ mm to 143.02 mm at $X_0 = 120$ mm, while the line-sensor error increased by 61.4% from 104.41 mm to 168.50 mm over the same range. Mean RMS error across the five X_0 values was 113.6 mm for the bump-sensors and 126.9 mm for the line sensors, indicating consistently lower errors for the bump sensor configuration. Both sensors degraded at comparable rates as X_0 increased, and errors were substantially higher in the arcing scenario than in the straight-line scenario for their respective X_0.

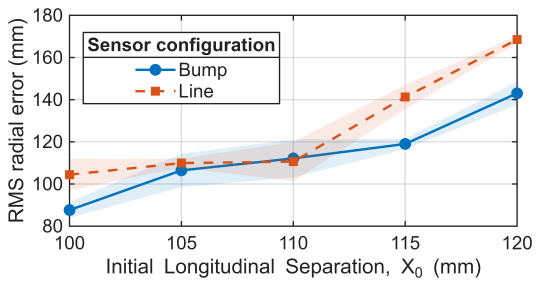


Fig. 5: Comparison of RMS radial error for the bump and line sensor configuration on the arcing path. Shaded regions represent ± 1 standard deviation across the 10 repeated trials.

C. Straight Path Trial

Figure 6 shows the mean RMS error results from the straight line trial, overlaid with derived uncertainty from the variation between the 10 tests.

The bump-sensor configuration produced consistently lower RMS radial error than the line sensors, while line-sensor error decreased with increasing X_0. Averaged across five lateral offsets, the line-sensor error fell by 56.8% between $X_0 = 100$mm and $X_0 = 120$mm while the bump-sensor error changed by only 1.8% over the same range. The spread of error across the lateral offsets also narrowed with increasing X_0, decreasing by 30.5% for the bump sensors (13.4mm to 9.3mm) and by 95.7% for the line sensors (36.7mm to 1.9mm). Although the bump-sensors maintained a narrower spread at each X_0 across all Y_0, the line sensors exhibited a steeper reduction in variability.

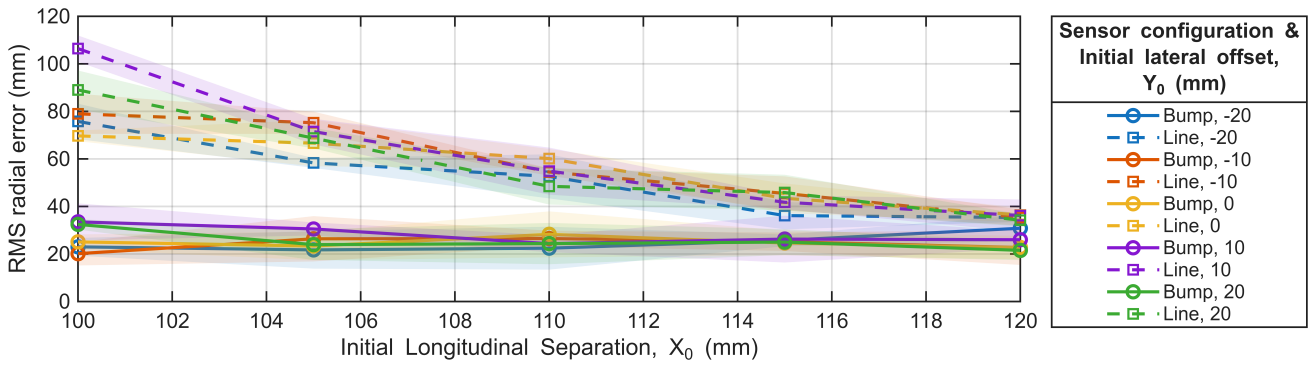


Fig. 6: Comparison of RMS radial error for the bump and line sensors with initial lateral and longitudinal offset on a straight path. Shaded regions represent ± 1 standard deviation across 10 repeated trials.

V. Discussion and Conclusion

As the initial longitudinal separation, lateral offset, and trajectory curvature increase, tracking error in leader-follower operation will increase. Specifically, bump sensor-based tracking is expected to achieve lower longitudinal error at larger separations, whereas line sensor-based tracking is expected to achieve lower lateral and overall positional error under lateral offsets and curved trajectories.

For the straight line trials, the observed reduction in RMS radial error with increased longitudinal separation refuted the original hypothesis. However, this does not indicate that larger separations intrinsically improved tracking performance. The controller functionality outlined in Algorithm 1 was intended to stop the follower once the received signal recovered to the level taken during calibration, such that all runs would terminate at approximately their starting longitudinal offset. In practice, this behaviour was not achieved exactly, and the follower frequently overshot before stopping. Consequently, runs with larger initial separation often involved a greater effective chase distance than intended, allowing initial transients to decay and for the RMS radial error to stabilise further before termination. Therefore, the lower error observed at larger X_0 in straight-line motion is more plausibly attributed to imperfect stop regulation and extended settling distances than to an inherent reduction in sensing difficulty.

The arced trials provided partial support for the hypothesis, highlighting that sensor performance depends on trajectory-induced sensing geometry rather than coverage alone, and should be distinguished between straight and curved motion. For both sensor configurations, RMS radial error increased approximately linearly with X_0 as expected. However, bump sensors outperformed line sensors in 80% of the trials, indicating that the expectation that the line-sensor configuration would perform better on curved trajectories was not supported. This suggests that for the curved trajectory used, the wider lateral sensing region of the line sensors did not outweigh their shorter forward range or dependence on adequate surface reflectance. Under arcing, the bump sensors likely benefited from a more stable line-of-sight signal, whereas the line sensors' reliance on reflected IR introduced greater sensitivity to angular misalignment and surface-dependent signal attenuation. The substantially greater errors observed in the curved-path trials compared to the straight-line trials also indicate that the more complex trajectory was a major source of tracking difficulty, with bearing errors accumulating throughout motion. Odometry-based position estimates also degrade with distance travelled due to the accumulation of incremental motion errors, while confidence in these estimates correspondingly decreases as displacement increases [4]. These results suggest that for curved motion, signal stability was more important than wider sensor coverage alone.

Overall, this investigation showed that the IR sensor configuration has a clear effect on the leader-follower tracking performance of Pololu 3Pi+ robots. Although both sensing approaches enabled closed-loop following on different trajectories, the bump sensor configuration produced lower RMS radial error than the line sensor configuration in both straight-line and curved-path trials. This indicates that simple IR sensing can support relative position tracking, but performance depends heavily on sensor orientation and trajectory for consistent signal strength. An important limitation was inconsistent ambient lighting throughout robot motion and its resulting effect on sensor readings. Additionally, the single controllers used for distance and bearing correction may have limited the maximum performance achievable for each sensor configuration. Future work should improve lighting consistency during trials by performing them in controlled environments. The tuning of heading correction and sensor sensitivity should also be investigated to determine their effects on performance.

In practice, leader-follower system design must match sensor characteristics and controller behaviour to achieve repeatable performance within an intended operating environment and region. Single controllers have an inherent optimal region, and methods such as gain scheduling could extend tracking across a wider envelope. Additionally, the usage of certain sensor topologies may not be applicable or reliable across all environments, which must be considered in behaviour design.

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